MIL-C-43748A
1 February 1978
SUPERSEDING
MIL-C-43748
9 August 1971

MILITARY SPECIFICATION

CUTTER, PAPER, GUILLOTINE; 12-INCH (304.8 mm) CAPACITY, BENCH MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size paper cutter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command (GL), Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

FSC 3610

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

MIL-L-2104 - Lubricating Oil, Internal Combustion Engine (Heavy-Duty)
MIL-P-3684 - Printing, Duplicating and Book Binding Equpment, Packaging Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-13-2424 - Paper Cutter, 12-Inch (304.8 mm) Bench Mounted; Items List

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2, and 6.3).

3.2 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part or model number or equal shall be functionally equal, of equal or better quality with the manufacturers part identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacturer of the first article or, if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 Material and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification (see 6.4).

3.4 Design and construction. The design and construction of the paper cutter shall be as specified herein, and as shown on the drawings.

3.4.1 Electrical system. Wiring shall conform to the requirements of NFPA Standard No. 70. A single harness shall be provided for all the wires within the paper cutter, and shall be securely bundled. The electrical system (including the harness) shall be moisture and fungus resistant treated with material conforming to and applied as specified in MIL-T-152. The insulation of electrical circuits shall withstand a dielectric test of twice the rated voltage plus 1,000 volts, without flashover or break down (see 4.5.1).

3.4.2 Hydraulic system. The hydraulic system shall be filled to operating capacity with fluid conforming to grade 10 of MIL-L-2104; and shall not leak when tested as specified in 4.5.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

3.5 Performance.

3.5.1 Operating controls. The cutting cycle shall be activated by nonrepeat, two-hand, interlocking, push-button control switches. The switches shall require the operator's two hands to start and continue the cutting stroke. Removal of either hand from the switches shall cause the cutting stroke to immediately stop and the knife to return to the start position (see 4.5.2).

3.5.2 Clamp bar. The clamp bar shall automatically clamp the paper pile when the cutting knife starts to cut the paper and shall continue to clamp the paper pile until the knife completes the cut and the cutting edge of the knife passes the bottom of the clamp bar on its return stroke (see 4.5.2.1).

3.5.3 Overloads. Built in overload protection shall be provided to automatically stop the down travel of the cutter bar when a sharp, previously unused knife strikes a wood object (see 4.5.2.3).

3.5.4 Paper cutting range. The paper cutters shall cut 9 pound (33.8 g/m^2) manifold, 50 pound (74 g/m^2) book, and 90 pound (162.7 g/m^2) index paper for not less than 10 minutes, when stacked from 2 to 3 inches (50.8 to 76.2 mm) high, without leakage of hydraulic fluid (see 4.5.2.2).

3.5.5 Paper cutting accuracy and quality. When tested as specified in 4.5.2.4 the cut sheets shall not vary in dimensions more than 1/16 inch (1.6 mm). The actual trimmed dimension shall correspond with the gauge reading within 1/32 inch (.8 mm). The adjacent edges of folded sheets shall be parallel within 1/32 inch (.8 mm). The trimmed edge of each pile shall be smooth and free of chatter marks. The individual sheets shall have straight and clean edges. All sheets within a pile shall be clean and free of oil stain and any other foreign matter.

3.5.6 Sustained performance. The paper cutter shall be capable of cutting two hundred and fifty thousand, 10-1/2 x 16-inch (266.7 x 406.4 mm) sheets of 20 pound (50 g/m^2) basis bond paper into 5-1/4 by 8-inch (133.3 x 203.2 mm) sheets in not more than 4 consecutive hours of operation (see 4.5.2.5). Any down time required for lubrication, changing of cutting blades or other nonrecurring maintenance, which can be accomplished within 5 minutes, shall be included within the 4 hours of operation.

3.5.7 Disassembly and assembly. After 4 hours of operation, the paper cutter shall be disassembled. There shall be no evidence of excessive or uneven wear of bearing surfaces. After complete reassembly the paper cutter shall be tested for not more than 2 hours for conformance to the requirements for paper cutting accuracy and quality (see 3.5.5). During the 2 hour period one hundred and twenty-five thousand sheets of 10-1/2 x 16-inch (266.7 x 406.4 mm) sheets of 20 pound (50 g/m^2) basis bond paper shall be cut into 5-1/4 x 8-inch (133.3 x 203.2 mm) leaflets. Any down time for nonrecurring maintenance, which can be accomplished within 5 minutes, shall be permitted and included within the 2 hour period (see 4.5.2.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

3.6 Electromagnetic compatability. The end item shall be designed and equipped to conform to class IIB of MIL-STD-461 (see 4.5.2.7).

3.7 Contract data. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.8 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.9 Workmanship. The cutter shall be free from defects, splits, punctures, tears, dents, creases, deteriorations, or malformations. The finish shall be continuous, smooth, adherent, have no imbedded foreign matter and contain no sage, runs, drips, creeps, laps, bubbles, or streaks. No rust or tool marks shall show through the coating. Threaded fasteners shall not be broken, stripped, fractured or loose. All castings shall be free of blowholes or cracks.

3.9.1 Welding. The surfaces of parts to be welded shall be free from oxide, paint, grease and other foreign matter. Welds shall be sound continuous, smooth and free from porosity, cracks, incomplete fusion and deformation of material. All scale and flux deposits shall be removed from the finished welds.

3.9.2 Electrical wiring. Wiring shall not be cut, abraded or have excessive insulation stripped and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, the cutter shall be examined as specified in 4.4.2.1 and 4.4.2.2 and tested as specified in 4.5.1 thru 4.5.2.7. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 End item inspection. The lot shall be all paper cutters offered for inspection at one time. The sample unit for this examination shall be one assembled paper cutter including spare parts and tools.

4.4.2.1 Visual examination. Examination shall be made of the end item for defects in table I. The inspection level shall be II and the AQL shall be 2.5 for major defects, and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished where required	X	
	Finish not as specified	X	
	Not continuous, smooth, adherent, foreign matter imbedded and sags, runs, drips, creeps, laps, bubbles or streaks		X
	Rust or tool marks		X
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	X	
	Component fractured, split, punctured, torn, dented, creased, deteriorated or malformed	X	
Castings	Blowholes or cracks	X	
Welding	Missing, incomplete, burn holes, cracked, fractured, not fused	X	
	Undercut, not smooth and uniform or scale or flux deposits not removed	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other threaded fasteners	Missing, broken, stripped or fractured	X	
	Threaded fasteners loose		X
Electrical	Wiring cut, abraded, insulation stripped, not properly joined or loose at terminal	X	
	Adequate slack not provided for wiring to relieve strain	X	
Marking for identification	Missing, or illegible		X

4.4.2.2 Dimensional examination. Examination shall be made of the cutter for compliance with dimensions specified. Any dimension that is not within the specified tolerance shall be classified as a defect. The inspection level shall be S-2, with an AQL of 4.0, expressed in terms of defects per hundred units.

4.4.2.3 Testing of the end item. When electromagnetic compatability is required, and the applicable test was not accomplished under 4.3, the first produced end item shall be tested as specified in 4.5.2.7. In addition, each end item shall be tested as specified in 4.4.1 thru 4.5.2.6. Failure to pass either test shall be cause for reject the effected end item.

4.4.3 Packaging inspection. Preservation-packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in MIL-P-3684.

4.5 Tests.

4.5.1 Dielectric strength. The insulation of all electrical circuits shall be subjected to twice the rated voltage, plus 1,000 volts. The test potential may be direct current or peak alternating current of the same value, and shall be applied for 1 minute between all wiring and adjacent conductors not in the same circuit, and between all wiring and ground (see 3.4.1). Any evidence of flashover or breakdown shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

4.5.2 Performance test, operating controls.

(a) With the cutter switch on, alternatively depress and release the left and right hand push buttons at least six times. Commencement of the cutting stroke shall constitute failure of this phase of the test (see 3.5.1).

(b) The left hand push button shall be depressed and held. The right hand push button shall then be depressed. Both buttons shall remain depressed for not less than 10 seconds. Cycling of the cutting stroke more or less than once shall constitute failure of this phase of the test. This test shall then be repeated by depressing the buttons in the reverse order and observing, as before, for compliance with the nonrecycling requirements (see 3.5.1).

(c) The test in (b) above, shall be repeated except that the second button depressed shall be released prior to bottoming of the cutting knife. Failure of the knife to return to the start position, prior to bottoming, shall constitute failure of this phase of the test (see 3.5.1).

4.5.2.1 Clamp bar, test. During conduct of test 4.5.2 the bar clamp shall be observed. Failure of the bar clamp to automatically hold the paper during the operating cycle of the cutter shall constitute failure of this test (see 3.5.2).

4.5.2.2 Paper cutting range test. Paper, 9 pound (33.8 g/m^2), manifold, 50 pound (74 g/m^2) book, and 90 pound (162.7 g/m^2) index and stacked in piles 2 to 3 inches (50.8 to 76.2 mm) high, shall be continuously cut to random sizes for not less than 10 minutes. Any nonconformance with the cutting range requirements (see 3.5.4) or any evidence of leakage of the hydraulic system (see 3.4.2) shall constitute failure of this test.

4.5.2.3 Overloading test. One piece of knot free, straight grained, 6 to 12 inches (152.4 to 304.8 mm) long, framing lumber (hemlock, fir or spruce), having nominal dimensions of 2 x 4 inches (50.8 x 101.6 mm), shall be placed midway under the cutting knife. Both push buttons shall then be held depressed. A return of the cutting knife to the start position, before releasing the push buttons shall constitute failure of this test (see 3.5.3).

4.5.2.4 Cutting range and accuracy test. A pile of 10-1/2 x 16-inch (266.7 x 406.4 mm), 9 pound (33.8 g/m^2) manifold paper, stacked 3 inches (76.2 mm) high, shall be cut to 5-1/4 x 8-inch (133.4 x 203.2 mm) size. The paper piles shall be placed against the back gage and alternately against the left hand side gage and right hand side gage for succeeding cuts. The above procedure shall be repeated using 50 pound (74 g/m^2) book, and 90 pound (162.7 g/m^2) index paper. Sheets of paper at the bottom, middle and top of each 5-1/4 x 8-inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

(133.4 x 203.2 mm) pile shall be measured and compared with the gage readings and dimensional tolerance requirement in 3.5.5. The opposing 5-1/4 inch (133.4 mm) edges of each sample sheet shall be carefully aligned and the sheets folded to determine that all adjacent edges are parallel within 1/32 inch (.8 mm). Individual sheet edges, paper pile edges, and cleanliness shall also conform to the quality requirements in 3.5.5. Any paper cut with a variance more than 1/16 inch (1.6 mm), or not within 1/32 inch (.8 mm) of the gage reading, or not parallel within 1/32 inch (.8 mm) when folded shall constitute failure of this test.

4.5.2.5 Sustained performance test. The machine shall cut-two hundred and fifty thousand sheets of 10-1/2 x 16-inch (266.7 x 406.4 mm) 20 pound (50 g/m^2) basis bond paper to 5-1/4 x 8-inch (133.4 x 203.2 mm) size in not more than 4 hours of operation. Replacement of the cutting knife or nonrecurring maintenance or adjustments which can be accomplished within 5 minutes, shall be permitted within the test period. Any nonconformance with the paper cutting accuracy and quality (see 3.5.5) or stoppage in excess of 5 minutes, recurring maintenance adjustments (see 3.5.6) shall constitute failure of this test.

4.5.2.6 Disassembly and reassembly test. The paper cutter tested under 4.5.2.5 shall be disassembled, examined for uneven bearing surface wear, and then reassembled. After reassembly, one hundred and twenty five thousand sheets of 10-1/2 x 16-inch (266.7 x 406.4 mm) 20 pound (50 g/m^2) basis bond paper shall be cut to 5-1/4 x 8-inch (133.4 x 203.2 mm) size in not more than 2 hours. Replacement of the cutting knife or nonrecurring maintenance or adjustments which can be accomplished within 5 minutes, shall be permitted within the test period. Any evidence of excessive or uneven wear, stoppage in excess of 5 minutes, recurring maintenance adjustment (see 3.5.7) or non-conformance with paper cutting accuracy and quality (see 3.5.5) shall constitute failure of this test.

4.5.2.7 Electromagnetic compatability test. When electromagnetic compatability is required the first article or initial production unit, as applicable, shall be tested by the contractor in accordance with Test Method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461 (see 3.6).

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Preservation-packaging, packing, and marking for level A, B, or C, shall be in accordance with MIL-P-3684 (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The paper cutters are intended for use in mobile and printing plants for special warfare.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.2 and 6.3).
(c) Contract data requirements (see 3.7 and 6.2.1)
(d) Selection of applicable levels of preservation-packaging and packing (see 5.1).

6.2.1 Contract data requirements. Any data item to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulation of the procuring activity (see 3.7 and 6.2).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of the Armed Services Procurement Regulation (ASPR) 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.5 Metric equivalents. Metric equivalents, indicated in parenthesis throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA

Review activity:

DLA - GS

Preparing activity:

Army - GL

Project No. 3610-0247

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE	
MIL-C-43748A	CUTTER, PAPER, GUILLOTINE; 12-INCH (3.4.8 mm) CAPACITY, BENCH MOUNTED

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Natick Research and
Development Command
ATTN: DRXNM-TD
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:12Z
Check the source to verify that this is the current version before use.